

Adarsh Anand

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About

Backend engineer at Oracle building distributed cloud systems for identity governance and large-scale data processing. Experience in performance optimization, infrastructure automation, and microservice deployments across Kubernetes environments handling terabyte-scale data pipelines.

Education

Indian Institute of Technology Kanpur

B.Tech in Electrical Engineering

Kanpur, Uttar Pradesh

2019 – 2023

Work Experience

Oracle

Hyderabad, India

Member of Technical Staff IC2 – Access Governance

Jun 2023 – Present

- Mapped cross-module requirements for an **interconnected Access Governance dashboard** (entitlements, roles, user activity), building **40+ SQL queries** with complex joins across **18 tables / 100+ indexes**; reduced query latency by **6.5×** via AWR-guided tuning and partitioning.
- Developed automated shell scripts to deploy Docker containers of **50+ microservices** across **15+ Kubernetes namespaces** via Helm charts, reducing deployment time by over **60%**. (*DevOps, Kubernetes, Helm, Zsh*)
- Optimized logging system by removing **39% redundant data** (12GB → 7.3GB/day) using regular expressions, and improved processing speed by **400%** through Helm-based log truncation. (*OCI Logging, Regex, Groovy*)
- Transformed **10TB+ legacy data** using JOLT specs from **30+ targets** and Parquet formatting, implemented **Redis caching** to achieve **65% API latency reduction** with **95% cache hit rate**. (*ETL, Redis, Parquet*)
- Engineered a **critical Direct Modify Account feature** by consuming TMS APIs and merging API responses with static configuration and database data to construct **JSON Schema-compliant payloads** enabling UI-level editing of complex nested attributes and resolving a 3 year old technical debt. (*REST APIs, JSON Schema*)
- Led **end-to-end development** of Oracle Analytics Cloud (OAC) **POC reports** for CorpIDM onboarding under Audit & Compliance requirements. (*OAC, SQL*)
- Defined **report architecture and development guidelines** adopted by the team, delivered **knowledge transfer sessions to 4 senior engineers**, and collaborated directly with the **Chief Architect** to finalize requirements and report design. (*Audit & Compliance*)

Oracle

Bangalore, India

Intern – Oracle Content Management (OCM)

May 2022 – Jul 2022

- Developed an algorithm to prevent **Cross Site Scripting (XSS) attacks** on files uploaded on the OCM system.
- Received a **Pre-Placement Offer** from Oracle for exemplary performance in the Internship at the OCM division.

Leadership Experience

Placement Office, IIT Kanpur

Kanpur, Uttar Pradesh

Placement Coordinator

Mar 2021 – Feb 2022

- Led 70+ company coordinators & 100+ volunteers to execute IIT Kanpur's internship drive for 900+ students, architecting a unified Career portal for 1400+ candidates and reducing administrative overhead by 40%.
- Cultivated strategic partnerships with 50+ Fortune 500/GCC firms (Google, Microsoft, Optiver) and built a corporate database tracking 1000+ companies through stakeholder negotiations and campus recruitment events.

Key Projects

Local AI Environment – M

Personal Project | Open WebUI, LM Studio, Gemma 4

2026

- Deployed a **local agentic workflow** using **Docker** containers, integrating **Open WebUI** with **LM Studio** for private, offline LLM orchestration.
- Hosted **Gemma 4** locally using **Q_4_K_M quantization** to drastically reduce VRAM footprint while maximizing inference speed and preserving reasoning limits.

Gold-Denominated World Assets GDWA – G

Personal Project | Python, Streamlit, Plotly, Pandas, yFinance

2025

- Built a **financial analytics dashboard** to benchmark **10+ global assets** (S&P 500, Nifty 50, Nikkei, BTC, ETH, Silver) against Gold (XAU), stripping fiat inflation to reveal true purchasing power performance.
- Engineered **multi-currency denomination engine**, **incremental data sync**, and interactive analytics including **rolling 30-day volatility**, **underwater drawdown heatmaps**, and **asset correlation matrices**.

Algorithmic Trading & Backtesting Suite –

Personal Project | Python, Golang, Jupyter, Binance API

2024

- Developed a **full-stack trading toolkit** comprising a **strategy backtester**, **live crypto mock-trading engine**, **real-time market data fetcher** via Binance WebSocket, and an **AI-model-driven strategy** module.
- Built a **high-performance live data pipeline** in **Golang** with order book tracking and a Python layer for strategy execution, signal generation, and performance analytics across equities and crypto.

Stock Prediction using Reinforcement Learning

Finance and Analytics Club, IIT Kanpur

Jun 2021 – Jul 2021

- Developed and compared multiple ML models like SVM, KNN, and Regression to predict stock price movements.
- Designed a Reinforcement Learning agent based on policy gradient and epsilon greedy strategy concepts.
- Implemented action-reward based Deep Q-Learning networks using PyTorch to optimize trading decisions.

Mathematical Finance and Stochastic Processes

Stamatics, IIT Kanpur

Apr 2021 – May 2021

- Studied different option pricing methods and the Black-Scholes Model for option pricing.
- Analyzed risk and return of stocks and their relationships using Modern Portfolio Theory.
- Eliminated risk to obtain optimal weights for maximum Sharpe ratio to design Markowitz efficient frontier.

Tweet Sentiment Analysis

Course Project – Data Mining and Knowledge Discovery

Sep 2021 – Dec 2021

- Applied pre-processing on about 2 million data points to perform lemmatization and stemming on tweets.
- Used NLP-based word embedding techniques including TF-IDF and GloVe methods.
- Employed SVM and LSTM models to achieve an accuracy of 78% and F1 score of 0.82.

Technical Skills

Languages: C, C++, Java, Python, Golang, JavaScript, PL/SQL

Backend & Distributed Systems: REST APIs, Microservices, Redis, Kafka, Flink

Infrastructure: Kubernetes, Docker, Helm, CI/CD

Extracurricular Activities

- * **Toastmasters International (IIT Kanpur):** Enhanced public speaking and leadership skills of 100+ students over 20+ GDs, impromptu and speech sessions as one of the founding members of the first Toastmasters IIT Kanpur Chapter.

Relevant Courses

Fundamentals of Computing • Data Structures and Algorithms • Stochastic Processes and their Applications
 Financial Engineering • Linear Algebra and Differential Equations • Discrete Mathematics
 Probability and Statistics • Equity Valuation and Derivative Contracts • Micro and Macroeconomics